

ASSIGNMENT-9

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BATCH: 30

QUESTION:

Interacting with DeFi Protocols (Testnet)

Objective

To interact with a decentralized finance (DeFi) protocol on a blockchain testnet by writing and executing a smart contract that deposits and withdraws tokens, demonstrating basic DeFi operations without real funds.

Requirements

- Install Node.js and npm
- Install VS Code
- Install VS Code extensions:
 - o Solidity
 - o Hardhat
- Use a testnet (Sepolia / Goerli / Mumbai)
- Create a smart contract and a script to interact with a DeFi protocol (mock lending contract)
- Use functions for deposit and withdrawal
- Test interactions using test tokens
- Add inline comments explaining the logic

Tools & Technologies

- Solidity
- Hardhat
- JavaScript (Node.js)
- Ethereum Testnet (Sepolia)

- MetaMask (Testnet account)

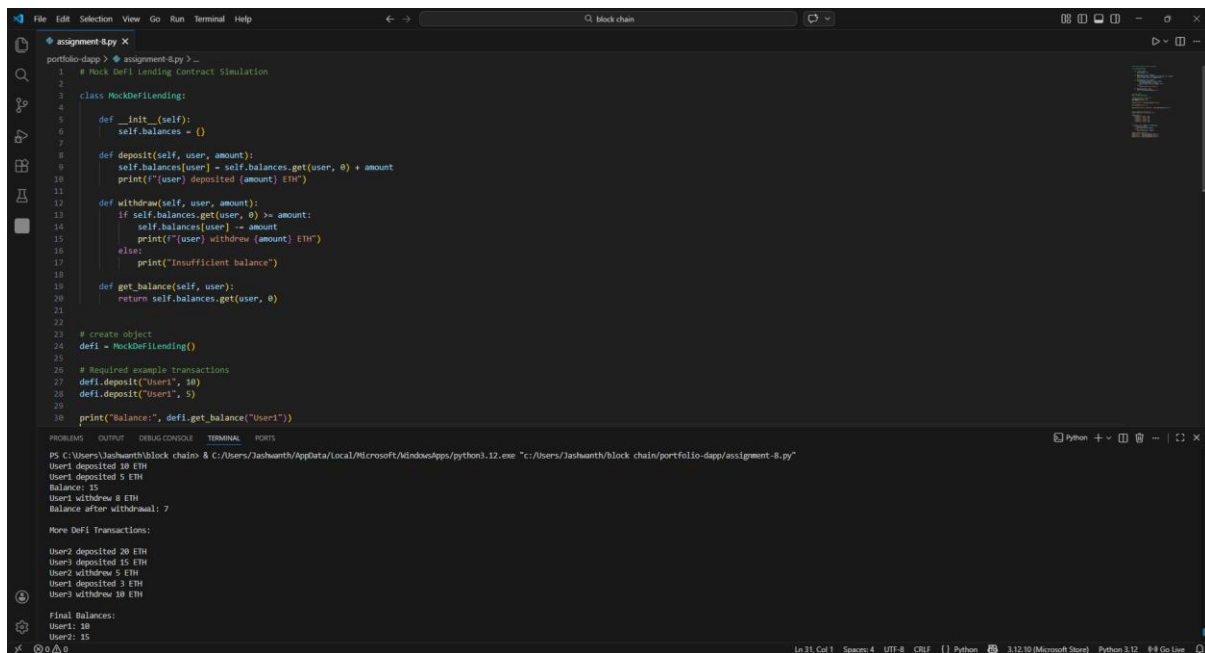
Practical Description

In this practical, we simulate interaction with a DeFi protocol by creating a simple lending contract where:

- Users can deposit ETH
- Users can withdraw ETH
- Contract tracks balances internally

This represents how DeFi protocols like Aave or Compound work on testnets.

CODE:



```
1 # Mock DeFi lending Contract Simulation
2
3 class MockDeFiLending:
4
5     def __init__(self):
6         self.balances = {}
7
8     def deposit(self, user, amount):
9         self.balances[user] = self.balances.get(user, 0) + amount
10        print(f"{user} deposited {amount} ETH")
11
12    def withdraw(self, user, amount):
13        if self.balances.get(user, 0) >= amount:
14            self.balances[user] -= amount
15            print(f"{user} withdrew {amount} ETH")
16        else:
17            print("Insufficient balance")
18
19    def get_balance(self, user):
20        return self.balances.get(user, 0)
21
22
23 # create object
24 defi = MockDeFiLending()
25
26 # Required example transactions
27 defi.deposit("User1", 10)
28 defi.deposit("User1", 5)
29
30 print("Balance:", defi.get_balance("User1"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Jashwanth\block chain> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "C:\Users\Jashwanth\block chain\portfolio-dapp\assignment-8.py"

User1 deposited 10 ETH
User1 deposited 5 ETH
Balance: 15
User1 withdrew 8 ETH
Balance after withdrawal: 7

More DeFi Transactions:

User2 deposited 20 ETH
User3 deposited 15 ETH
User2 withdrew 5 ETH
User1 deposited 3 ETH
User3 withdrew 10 ETH

Final Balances:
User1: 20
User2: 15

Ln 31, Col 1 Spaces: 4 UTF-8 CRLF Python 3.12.10 (Microsoft Store) Python 3.12 #4 Go Live